

Binomial Theorem related Proofs — Problem Set

Note: While the focus of this problem set is proofs using the Binomial Theorem, some questions are solved/proved using factorial notation

Warm-up

1. Simplify the expression $\frac{(n+1)!}{n!}$ in terms of n .
2. Write out the first four terms of the expansion $(1+x)^n$ in sigma notation.
3. Show that $\binom{n}{r} = \binom{n}{n-r}$ using factorial notation.
4. Show that $r\binom{n}{r} = n\binom{n-1}{r-1}$ by using the factorial definition.
5. Given the expansion $(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + \binom{n}{n}x^n$, what single value of x proves that the sum of coefficients is 2^n ?
6. Determine the value of $\binom{5}{0} - \binom{5}{1} + \binom{5}{2} - \binom{5}{3} + \binom{5}{4} - \binom{5}{5}$ by substituting $x = -1$ into $(1+x)^5$.
7. What value of x should be substituted into $(1+x)^{10}$ to evaluate the sum $\sum_{r=0}^{10} \binom{10}{r} 2^r$?
8. Find the value of $\sum_{k=0}^4 \binom{4}{k} 10^k$.
9. Evaluate $\binom{8}{0} - \binom{8}{1} + \binom{8}{2} - \cdots + \binom{8}{8}$.
10. Given $(2+x)^n = \sum_{r=0}^n \binom{n}{r} 2^{n-r} x^r$, what is the sum of the coefficients when $x = 1$?
11. Identify the binomial $(1+x)^n$ and the value of x represented by the series $1 + 3\binom{n}{1} + 9\binom{n}{2} + \cdots + 3^n\binom{n}{n}$.

Skill-building

1. Using an appropriate binomial expansion, state the value of $\binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \cdots + (-1)^n \binom{n}{n}$.
2. Find the coefficient of x^2 in the expansion of $(1+x)^5(1+x)^2$.
3. Show that $\frac{1}{r+1}\binom{n}{r} = \frac{1}{n+1}\binom{n+1}{r+1}$ for $0 \leq r \leq n$.
4. By considering the expansion of $(1+x)^n$, prove that $\sum_{k=0}^n 3^k \binom{n}{k} = 4^n$.
5. Show that $\binom{n}{r} \times \binom{r}{k} = \binom{n}{k} \times \binom{n-k}{r-k}$ for integers $n \geq r \geq k \geq 0$.
6. Prove that $\binom{2n}{2} = 2\binom{n}{2} + n^2$ by using the factorial formula.
7. Find the coefficient of x^n in the product $(1+x)^n(1+x)^n$ and explain why this equals $\sum_{r=0}^n \binom{n}{r}^2$.

8. By expanding $(1 - x^2)^n = (1 - x)^n(1 + x)^n$, find an expression for $\sum_{k=0}^n (-1)^k \binom{n}{k}^2$ when n is odd.
9. Prove that $\binom{n+2}{r} = \binom{n}{r} + 2\binom{n}{r-1} + \binom{n}{r-2}$ for $r \geq 2$.
10. Consider the identity $(1 + x)^m(1 + x)^n = (1 + x)^{m+n}$. By equating the coefficients of x^r on both sides, prove that $\sum_{k=0}^r \binom{m}{k} \binom{n}{r-k} = \binom{m+n}{r}$.

Easier Exam Questions

1. (Glenwood 2022 Q9c)
Show that for all integers $n \geq 3$, **3**

$$\binom{n+2}{3} - \binom{n}{3} = n^2$$
2. (Presbyterian 2021 Q7c)
Show that ${}^{n+1}C_r \div {}^nC_{r-1} = \frac{n+1}{r}$. **2**
3. (Gosford 2020 Q10d)
i) Write down the binomial expansion of $(1 + x)^n$ **1**
 ii) Use the expansion of $(1 + x)^n$ to show that ${}^nC_0 + {}^nC_1 + \cdots + {}^nC_n = 2^n$ **2**
4. (Sydney Girls 2021 Q3a)
 Consider the binomial expansion: $(1 + x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \binom{n}{3}x^3 \cdots + \binom{n}{n}x^n$. **1**
 Find the value of: $\binom{100}{0} + \binom{100}{1} + \binom{100}{2} + \cdots + \binom{100}{100}$.
5. (Blacktown Boys 2023 Q10)
By considering the binomial expansion of $(1 + x)^{10}$, what is the value of

$$5 \times {}^{10}C_1 + 5^2 \times {}^{10}C_2 + 5^3 \times {}^{10}C_3 + \cdots + 5^{10} \times {}^{10}C_{10}$$

 (A) 4^{10}
 (B) 6^{10}
 (C) $4^{10} - 1$
 (D) $6^{10} - 1$
6. (Baulkham Hills 2019 Q14d)
 Using the fact that $(1 + x)^4(1 + x)^9 \equiv (1 + x)^{13}$, show that **2**

$$\binom{4}{0}\binom{9}{4} + \binom{4}{1}\binom{9}{3} + \binom{4}{2}\binom{9}{2} + \binom{4}{3}\binom{9}{1} + \binom{4}{4}\binom{9}{0} = \binom{13}{4}$$
7. (Sydney Girls 2022 Q2f)
 By using $(1 + x)^{10} = (1 + x)^5(1 + x)^5$, show that **2**

$$\binom{10}{3} = \binom{5}{3}\binom{5}{0} + \binom{5}{2}\binom{5}{1} + \binom{5}{1}\binom{5}{2} + \binom{5}{0}\binom{5}{3}.$$

8. (Glenwood 2023 Q10c)

By expanding both sides of the identity $(1+x)^{11} = (1+x)^8(1+x)^3$, Prove that :

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$$\binom{11}{3} = \binom{8}{0}\binom{3}{3} + \binom{8}{1}\binom{3}{2} + \binom{8}{2}\binom{3}{1} + \binom{8}{3}\binom{3}{0}$$

Harder Exam Questions

1. (Girraween 2024 Q11c)

i. Write down the binomial expansion of $(1+x)^n$.

1

ii. Hence, show that $\binom{n}{1} + \binom{n}{2} + \binom{n}{3} + \dots + \binom{n}{n} = 2^n - 1$.

2

iii. By substituting a suitable value for x in the expansion $(1+x)^n$, show that:

2

$$1 + 2\binom{n}{1} + 4\binom{n}{2} + \dots + 2^{n-1}\binom{n}{n-1} = 3^n - 2^n$$

2. (Girraween 2023 Q11 b,e, & f)

b. By considering $(1+x)^{11} = (1+x)^3(1+x)^8$, show that

3

$${}^{11}\mathbf{C}_5 = {}^8\mathbf{C}_5 + {}^3\mathbf{C}_1 {}^8\mathbf{C}_4 + {}^3\mathbf{C}_2 {}^8\mathbf{C}_3 + {}^8\mathbf{C}_2$$

.

e. By considering the binomial expansion

$$(1+x)^{2n} = {}^{2n}\mathbf{C}_0 + {}^{2n}\mathbf{C}_1 x + {}^{2n}\mathbf{C}_2 x^2 + \dots + {}^{2n}\mathbf{C}_n x^n + \dots + {}^{2n}\mathbf{C}_{2n} x^{2n}$$

show that:

3

$$2^{2n} = 2 \sum_{r=0}^n {}^{2n}\mathbf{C}_r - {}^{2n}\mathbf{C}_n$$

f. (i) By using binomial expansion, show that

2

$$(q+p)^n - (q-p)^n = 2 {}^n\mathbf{C}_1 q^{n-1} p + 2 {}^n\mathbf{C}_3 q^{n-3} p^3 + \dots$$

(ii) Find the last term in the expansion when n is odd.

1

(iii) Find the last term in the expansion when n is even.

1

3. (Sydney Girls 2019 Q4b)

By substituting a suitable value for x in the expansion $(1+x)^n$, show that

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$$1 + 2\binom{n}{1} + 4\binom{n}{2} + \dots + 2^{n-1}\binom{n}{n-1} = 3^n - 2^n.$$

4. (Baulkham Hills 2024 Q14c)

(i) Show that ${}^{n+1}C_r - {}^nC_r = {}^nC_{r-1}$, for $r = 1, 2, 3, \dots, n$.

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(ii) Hence find the value of ${}^3C_2 + {}^4C_2 + {}^5C_2 + \dots + {}^{100}C_2$.

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5. (Baulkham Hills 2025 Q14c)

i) Show that $x^n(1+x)^n\left(1+\frac{1}{x}\right)^n = (1+x)^{2n}$ **2**

ii) Hence show that $1 + \binom{n}{1}^2 + \binom{n}{2}^2 + \cdots + \binom{n}{n}^2 = 2^n C_n$ **3**

6. (Sydney Girls 2020 Q4d)

Suppose that x is a real number and that n is a positive integer. **2**

Show that

$$\cos^n 2x = \binom{n}{0} \cos^{2n} x - \binom{n}{1} \cos^{2n-2} x \sin^2 x + \binom{n}{2} \cos^{2n-4} x \sin^4 x - \cdots + (-1)^n \binom{n}{n} \sin^{2n} x$$

7. (Girraween 2022 Q8a)

(i) By expanding $(1+x)^n$ and using an appropriate substitution, show that ${}^nC_0 + {}^nC_1 + {}^nC_2 + {}^nC_3 + \cdots + {}^nC_n = 2^n$ **3**

(ii) By using an appropriate substitution in the expansion of $(1+x)^n$, show that ${}^nC_0 + {}^nC_2 + {}^nC_4 + \cdots = {}^nC_1 + {}^nC_3 + {}^nC_5 + \cdots$ **2**

8. (Glenwood 2019 Q21 a,b&c)

(a) Show that $(1-x)^5(1+\frac{1}{x})^5 = (\frac{1}{x}-x)^5$ **1**

(b) Write the expansion of $(\frac{1}{x}-x)^5$, leaving coefficients in the form $\binom{n}{r}$ **1**

(c) Determine the coefficient of x^3 in the expansion of $(1-x)^5(1+\frac{1}{x})^5$ and hence show that **2**

$$\binom{5}{4}\binom{5}{1} - \binom{5}{5}\binom{5}{2} - \binom{5}{3}\binom{5}{0} = \binom{5}{4}$$

9. (North Sydney Boys 2023 Section E Q2)

i) Show that $(1+x)^m\left(1-\frac{1}{x}\right)^m = \left(x-\frac{1}{x}\right)^m$ **1**

ii) By considering the term(s) independent of x in the expansion of the results from part i, show that: **3**

$$\binom{2002}{0}^2 - \binom{2002}{1}^2 + \binom{2002}{2}^2 - \cdots + \binom{2002}{2002}^2 = -1 \binom{2002}{1001}$$

iii) Hence, or otherwise, show that **3**

$$\sum_{k=0}^{1001} (-1)^k \binom{2002}{k}^2 = -\frac{1}{2} \binom{2002}{1001} [1 + \binom{2002}{1001}]$$